Licenciatura em Engenharia Informática
ALGAN
1st Semester 2025-2026
TP3

1. Consider the following systems of linear equations,

$$(\mathbf{S_1})\begin{cases} 2x_1+2x_3 = 1 \\ 3x_1-x_2+4x_3 = 7 \\ 6x_1+x_2+10x_3 = 0 \end{cases} \quad (\mathbf{S_2})\begin{cases} x_1+2x_2-x_4+x_5 = 1 \\ 3x_2+x_3-x_5 = 7 \\ x_4+7x_5 = 3 \end{cases} \quad (\mathbf{S_3})\begin{cases} x_1+2x_2-x_3 = 1 \\ x_2-x_3 = 7 \\ x_3 = 2 \end{cases}$$

   a) Determine the augmented matrix associated with each of the systems..
   b) Identify the matrices, found in the previous item, that are in row echelon form.
   c) Use the Gaussian elimination algorithm to solve the three systems..

2. Consider the following SLE in matricial form

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

   a) Prove that:

$$\mathbf{s} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{e} \quad \mathbf{s}' = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

   are two solutions of the SLE.
   b) Show that $\mathbf{s}'' = \mathbf{s} - \mathbf{s}'$ is a solution of the associated homogeneous system of linear equations.
   c) Show that $\mathbf{s} + \lambda\mathbf{s}''$ is solution of the SLE, for all $\lambda \in \mathbb{R}$.
   d) Based on the previous items, find an argument that supports the (true) proposition: "There are no systems of linear equations with a finite number of solutions greater than 1".

3. Consider the SLE

$$\begin{cases} x+y-3z+t = 0 \\ y-4z = 1 \\ x+2y-7z+t = 1 \\ x+z+t = -1 \end{cases}.$$

   a) Show that for all $\lambda \in \mathbb{R}$, $\mathbf{s} = (-1-\lambda, 1+4\lambda, \lambda, 0)$ is solution of the SLE.
   b) Determine whether the set $\mathcal{A} = \{(-1-\lambda, 1+4\lambda, \lambda, 0) : \lambda \in \mathbb{R}\}$ is the solution set of the system.

4. Classify, in terms of the parameters $\alpha, \beta \in \mathbb{R}$, the following systems of linear equations.

$$\mathbf{a)}\begin{cases} 3x_1-x_2+\alpha x_3 = 1 \\ 3x_1-x_2+x_3 = 5 \end{cases} \quad \mathbf{b)}\begin{cases} x_1+4x_2 = 2 \\ -3x_1+\alpha x_2 = -1 \end{cases} \quad \mathbf{c)}\begin{cases} 2x_1+5x_2-3x_3 = \alpha \\ 4x_1+7x_2-4x_3 = 0 \\ -6x_1-3x_2+x_3 = \beta \end{cases}$$

$$\mathbf{d)}\begin{cases} x_1+3x_2+x_3 = \alpha \\ -x_1-x_2+x_3 = \beta \\ 3x_1+7x_2-x_3 = \beta \end{cases} \quad \mathbf{e)}\begin{cases} x_1+2x_2-\alpha x_3 = 1 \\ 2x_1-x_2-x_3 = \beta \\ 9x_1-2x_2+x_3 = -1 \end{cases} \quad \mathbf{f)}\begin{cases} x+y+z = 3 \\ x+2y+(4-\alpha)z = 8+\beta \\ 2x+4y+(4-\alpha)z = 9-\beta \end{cases}$$

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5. Let $\mathbb{P}_3[x]$ be the set of all polynomials in variable x, with real coefficients and degree not exceeding 3. Determine, if it exists, a polynomial $p \in \mathbb{P}_3[x]$ that satisfies the following conditions:

   a) $p(-2) = p(-1) = p(1) = 1$ and $p(2) = 2$.

   b) $p(1) = 1$, $p(-3) = \frac{1}{3}$, $p'(1) = \frac{1}{2}$ and $p'(-3) = \frac{7}{6}$.

   c) $p + \frac{dp}{dx} = x^3$.

6. Consider the following matrices,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 5 & 2 & -1 \\ 0 & 1 & 3 & -4 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 1 & 4 & 2 & 3 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 \\ 1 & 4 & -4 & 1 & 2 \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 4 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 1 & 64 & 65 \\ 0 & 0 & 0 & 0 & 1 & 36 & 0 & 37 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 8 & 0 & 0 & 9 \end{bmatrix}$$

   a) Indique as matrizes que estão em forma de escada por linhas.

   b) Which matrices are in row echelon form?

   c) Calculate their rank.

   d) Assuming that the indicated matrices are augmented matrices associated with a system of linear equations, where the last column represents the constant term, classify the systems and find their solution sets.

7. **a)** Determine which of the following matrices are invertible.

   b) Determine the inverses by applying the Gauss-Jordan algorithm.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 & 6 \\ 1 & 4 & 6 & 12 \\ 1 & 4 & 8 & 16 \\ 1 & 2 & 4 & 8 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 3 & 9 & 0 & 0 \\ 3 & 9 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & -2 & -1 & 0 & 0 \\ 0 & -2 & 0 & 0 & -1 & 0 \\ -2 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} 4 & 8 & -7 & 14 \\ 2 & 5 & -4 & 6 \\ 0 & 2 & 1 & -7 \\ 3 & 6 & -5 & 10 \end{bmatrix}$$

8. Let α, β, δ, γ be non-zero real numbers. Determine the inverse of each of the matrices.

$$\mathbf{A} = \begin{bmatrix} \alpha & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 \\ 0 & 0 & \delta & 0 \\ 0 & 0 & 0 & \gamma \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & \alpha \\ 0 & 0 & \beta & 0 \\ 0 & \delta & 0 & 0 \\ \gamma & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} \alpha & 0 & 0 & 0 \\ 1 & \alpha & 0 & 0 \\ 0 & 1 & \alpha & 0 \\ 0 & 0 & 1 & \alpha \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} \alpha & 0 & 0 & 0 \\ 1 & \beta & 0 & 0 \\ 0 & 1 & \delta & 0 \\ 0 & 0 & 1 & \gamma \end{bmatrix}$$

9. Consider the following SLE.

$$(\mathbf{S_1})\begin{cases} x & & & + & z & = 1 \\ & & y & + & z & = -2 \\ 3x & + & 8y & - & 4z & = 1 \end{cases} \qquad (\mathbf{S_2})\begin{cases} x & + & 2y & + & 3z & = -2 \\ x & + & 4y & + & 9z & = -6 \\ x & + & 8y & + & 27z & = -20 \end{cases}$$

$$(\mathbf{S_3})\begin{cases} x & + & 2y & + & 3z & = -5 \\ 5x & + & y & + & 2z & = 1 \\ -x & & & - & 4z & = 6 \end{cases} \qquad (\mathbf{S_4})\begin{cases} x & + & y & + & z & + & t & = & 4 \\ x & + & 2y & + & z & + & t & = & 5 \\ x & + & 2y & + & 3z & + & t & = & 7 \\ 4x & & & + & 3z & & & = & 7 \end{cases}$$

a) Show that they are Cramer's systems.

b) Solve the systems using Cramer's formulas.

10. Consider the following systems of linear equations $\begin{cases} x + by + az = 1 \\ x + y + bz = a \\ x + by + bz = b \end{cases}$ where a e b are real parameters.

a) What condition must the parameters satisfy for the rank of the coefficient matrix of the SLE to be maximized?

b) Classify the SLE in terms of parameters a and b.

Solutions 1.

a) Denoting $[\mathbf{A_i}|\mathbf{b_i}]$ as the augmented matrix $(\mathbf{S_i})$, $i = 1, 2, 3$, we have:

$$[\mathbf{A_1}|\mathbf{b_1}] = \left[\begin{array}{ccc|c} 2 & 0 & 2 & 1 \\ 3 & -1 & 4 & 7 \\ 6 & 1 & 10 & 0 \end{array}\right] \quad [\mathbf{A_2}|\mathbf{b_2}] = \left[\begin{array}{ccccc|c} 1 & 2 & 0 & -1 & 1 & 1 \\ 0 & 3 & 1 & 0 & -1 & 7 \\ 0 & 0 & 0 & 1 & 7 & 3 \end{array}\right] \quad [\mathbf{A_3}|\mathbf{b_3}] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -1 & 7 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

b) $[\mathbf{A_2}|\mathbf{b_2}]$ e $[\mathbf{A_3}|\mathbf{b_3}]$.

c) $(\mathbf{S_1})$ consistent independent with set solution $CS_1 = \left\{\left(0, -5, \frac{1}{2}\right)\right\}$.

$(\mathbf{S_2})$ consistent dependent with degree 2) with set solution

$$CS_2 = \left\{\left(\frac{1}{3}(-2 + 2x_3 - 26x_5), \frac{1}{3}\left(7 - x_3 + x_5\right), x_3, 3 - 7x_5, x_5\right) \,:\, x_3, x_5 \in \mathbb{R}\right\}.$$

$(\mathbf{S_3})$ consistent independent with set solution $CS_1 = \{(-15, 9, 2)\}$.

2.

3.

b) No, the SLE has degree 2. In fact we have:

$$CS = \{(-1 - \lambda - \mu, 1 + 4\lambda, \lambda, \mu) \,:\, \lambda, \mu \in \mathbb{R}\}.$$

then $\mathcal{A} \subsetneq CS$.

4.

a) $\begin{cases} \alpha \neq 1 \text{ SCI.} \\ \alpha = 1 \text{ SI.} \end{cases}$

b) $\begin{cases} \alpha = -12 \text{ SI.} \\ \alpha \neq -12 \text{ SCI.} \end{cases}$

c) $\begin{cases} \beta - 5\alpha \neq 0 \text{ SI.} \\ \beta - 5\alpha = 0 \text{ SCD.} \end{cases}$

d) SCI, for all α, β.

e) $\begin{cases} \alpha = -5 \begin{cases} \beta = -\frac{1}{2}; \text{ SCD.} \\ \beta \neq -\frac{1}{2}; \text{ SI.} \end{cases} \\ \\ \alpha \neq -5, \ \forall \beta \in \mathbb{R} \text{ SCI, .} \end{cases}$

f) $\begin{cases} \text{If } \alpha \neq 4 \text{ tem-se SCD} \\ \text{If } \alpha = 4 \wedge \beta = -\frac{7}{3} \text{ tem-se SCI, Degree=1} \\ \text{If } \alpha = 4 \wedge \beta \neq -\frac{7}{3} \text{ tem-se SI} \end{cases}$

5.

a) $p(x) = \frac{5}{6} - \frac{1}{12}x + \frac{1}{6}x^2 + \frac{1}{12}x^3$.

b) $p(x) = \frac{5}{6} - \frac{1}{12}x + \frac{1}{6}x^2 + \frac{1}{12}x^3$.

c) $p(x) = -6 + 6x - 3x^2 + x^3$.

6.

a) **B** e **C**.

b) Denoting by $\mathbf{M}$ por $\tilde{\mathbf{M}}$ an row echelon form of a matrix, we have:

$$\tilde{\mathbf{A}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{\mathbf{B}} = \begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{\mathbf{C}} = \begin{bmatrix} 1 & 0 & -13 & 19 \\ 0 & 1 & 3 & -4 \end{bmatrix}$$

$$\tilde{\mathbf{D}} = \begin{bmatrix} 1 & 0 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad \tilde{\mathbf{E}} = \begin{bmatrix} 1 & 0 & 0 & -\frac{3}{5} & \frac{2}{5} \\ 0 & 1 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{5} & \frac{1}{10} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{\mathbf{F}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

c) $\mathrm{r}(\mathbf{A}) = 3$; $\mathrm{r}(\mathbf{B}) = 3$; $\mathrm{r}(\mathbf{C}) = 2$; $\mathrm{r}(\mathbf{D}) = 4$; $\mathrm{r}(\mathbf{E}) = 3$; $\mathrm{r}(\mathbf{F}) = 7$.

d)
- Associated system to $\mathbf{A}$, consistent dependent, with
$$SS = \left\{(0, x_2, 0, -x_2, x_5) \in \mathbb{R}^5 : \ x_2, x_5 \in \mathbb{R}\right\}.$$
- Associated system to $\mathbf{B}$, inconsistent
- Associated system to $\mathbf{C}$, consistent dependent, with
$$SS = \left\{(19 + 13x_3, -(4 + 3x_3), x_3) \in \mathbb{R}^3 : \ x_3 \in \mathbb{R}\right\}.$$
- Associated system to $\mathbf{D}$, consistent independent, with $SS = \{(-4, 5, -1, 1)\}$.
- Associated system to $\mathbf{E}$, consistent dependent, with
$$SS = \left\{\left(\frac{2}{5} + \frac{3}{5}x_4, \frac{1}{2} - x_4, \frac{1}{10} - \frac{3}{5}x_4, x_4\right) \in \mathbb{R}^4 : \ x_4 \in \mathbb{R}\right\}.$$
- Associated system to $\mathbf{F}$, consistent independent, with $SS = \{(1, 1, 1, 1, 1, 1, 1)\}$.

7.

a) They are all invertible.

b)

$$\mathbf{A^{-1}} = \begin{bmatrix} 0 & 0 & -1 & 2 \\ 0 & 1 & -1/2 & -1/2 \\ 1 & -1/2 & 1/2 & -1 \\ -1/2 & 0 & 0 & 1/2 \end{bmatrix} \quad \mathbf{B^{-1}} = \begin{bmatrix} 1/3 & 0 & 0 & 0 \\ -1/9 & 1/9 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -2/9 & 8/9 & -1 & 1 \end{bmatrix} \quad \mathbf{C^{-1}} = \begin{bmatrix} 1/2 & 1/4 & 1/4 & -1/4 \\ -1/2 & 1/4 & 1/4 & -1/4 \\ 0 & 1/2 & -1/2 & 1/2 \\ 0 & -1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\mathbf{D^{-1}} = \begin{bmatrix} -2/3 & 1/3 & 1/3 & 1/3 \\ 1/3 & -2/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & -2/3 & 1/3 \\ 1/3 & 1/3 & 1/3 & -2/3 \end{bmatrix} \quad \mathbf{E^{-1}} = \tfrac{1}{5}\mathbf{E} \quad \mathbf{F^{-1}} = \begin{bmatrix} 27 & -10 & 4 & -29 \\ -16 & 5 & -2 & 18 \\ -17 & 4 & -2 & 20 \\ -7 & 2 & -1 & 8 \end{bmatrix}$$

8.

$$\mathbf{A^{-1}} = \begin{bmatrix} \frac{1}{\alpha} & 0 & 0 & 0 \\ 0 & \frac{1}{\beta} & 0 & 0 \\ 0 & 0 & \frac{1}{\delta} & 0 \\ 0 & 0 & 0 & \frac{1}{\gamma} \end{bmatrix} \quad \mathbf{B^{-1}} = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{\gamma} \\ 0 & 0 & \frac{1}{\delta} & 0 \\ 0 & \frac{1}{\beta} & 0 & 0 \\ \frac{1}{\alpha} & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{C^{-1}} = \begin{bmatrix} \frac{1}{\alpha} & 0 & 0 & 0 \\ -\frac{1}{\alpha^2} & \frac{1}{\alpha} & 0 & 0 \\ \frac{1}{\alpha^3} & -\frac{1}{\alpha^2} & \frac{1}{\alpha} & 0 \\ -\frac{1}{\alpha^4} & \frac{1}{\alpha^3} & -\frac{1}{\alpha^2} & \frac{1}{\alpha} \end{bmatrix}$$

$$\mathbf{D^{-1}} = \begin{bmatrix} \frac{1}{\alpha} & 0 & 0 & 0 \\ -\frac{1}{\alpha\beta} & \frac{1}{\beta} & 0 & 0 \\ \frac{1}{\alpha\beta\delta} & -\frac{1}{\beta\delta} & \frac{1}{\delta} & 0 \\ -\frac{1}{\alpha\beta\delta\gamma} & \frac{1}{\beta\delta\gamma} & -\frac{1}{\delta\gamma} & \frac{1}{\gamma} \end{bmatrix}$$

9.

b) $CS_1 = \left\{(\frac{29}{15}, -\frac{16}{15}, -\frac{14}{15})\right\}$, $CS_2 = \{(-1, 1, -1)\}$, $CS_3 = \left\{(\frac{34}{35}, -\frac{13}{35}, -\frac{61}{35})\right\}$ e $CS_4 = \{(1, 1, 1, 1)\}$.

9.

a) $a \neq b \wedge b \neq 1$

b)
- If $a \neq b \wedge b \neq 1$ SCI
- If $a = b \neq 1$ SI
- If $a = b = 1$ SCD, with degree=2
- If $b = 1 \wedge a \neq 1$ SI